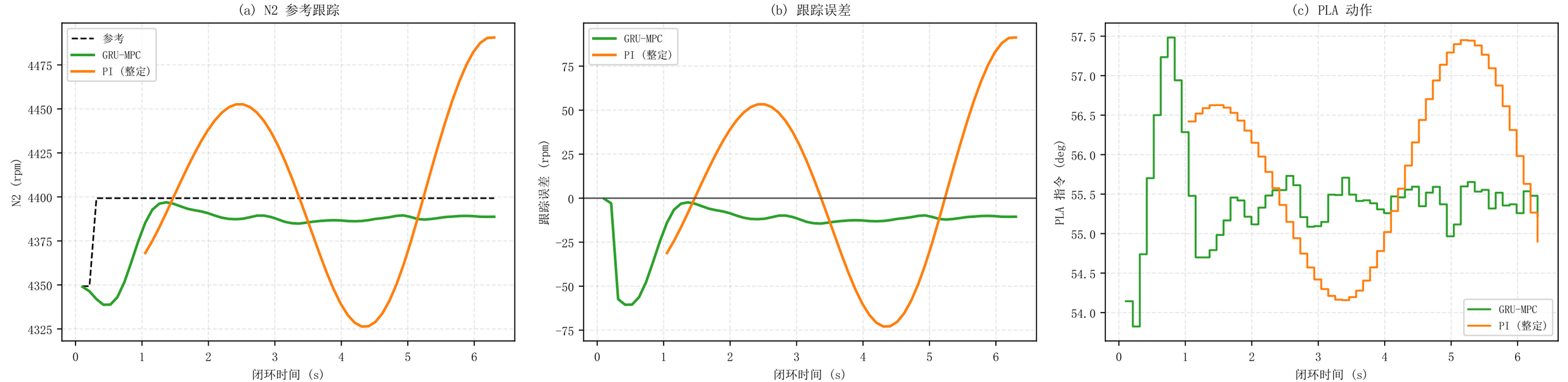图 1　真实 Simulink 闭环：GRU-MPC（60 步）vs PI 整定（51 步）